HOW-TO-GUIDE

demonstrating FA using the virtual Fabric Attach Demonstrator

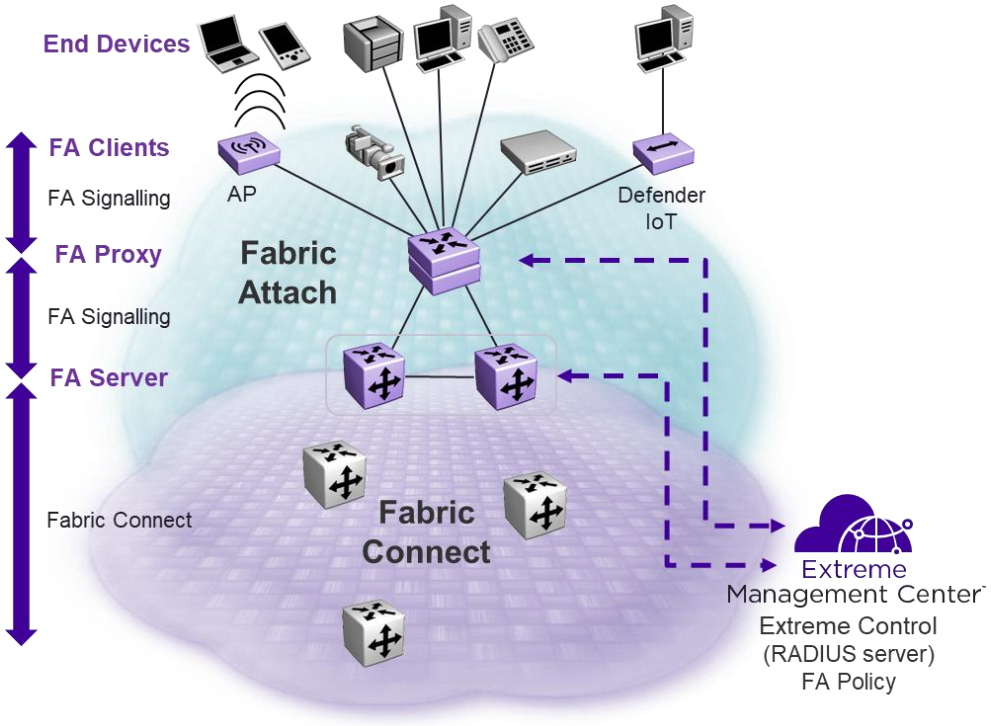
1. Target audience

System Engineers are interested in understanding the Fabric Attach (FA) technology for test and demo purposes. Why does it make sense to use FA? It's because it automates the configuration at the edge of the network. Shortest Path Bridging (SPB) already addresses the simplification in the Data Centre and Campus. The combination of SPB and FA is a perfect fit for a flexible, easily managed network access environment.

The idea of Fabric Attach in the Enterprise

Having simplified the network core and automated it with Fabric Connect (FC), Extreme Networks recognised that solving the attach problem in the campus with devices such as IP CCTV cameras or Wireless Access points would provide immense operational and business benefits.

Extreme Networks solved the problem by designing Fabric Attach (FA), a technology that enables an edge device, be it a virtual machine, am Fabric Engine, Switch engine, ISW, Open vSwitch, IP CCTV camera, Wireless Access point or even a legacy switch unable to support SPB to automatically attach to a virtual service network (VSN) without any human intervention.



Fabric Attach, known also as auto-attach as it becomes standardized in IEEE 802.1Qcj, uses simple LLDP messages for the device to request attachment to a specific L2 VSN. The Fabric Connect network with the Fabric Attach Server function then ensures the provisioning as needed.

Prerequisites

There are two choices available.

- Open Virtual Appliance (OVA) runnable under
  VMware ESX 6.x or later, VMware Workstation/Player and VirtualBox.
  The VM requires 1 GB RAM, 1 CPU core, and 8 GB disk space

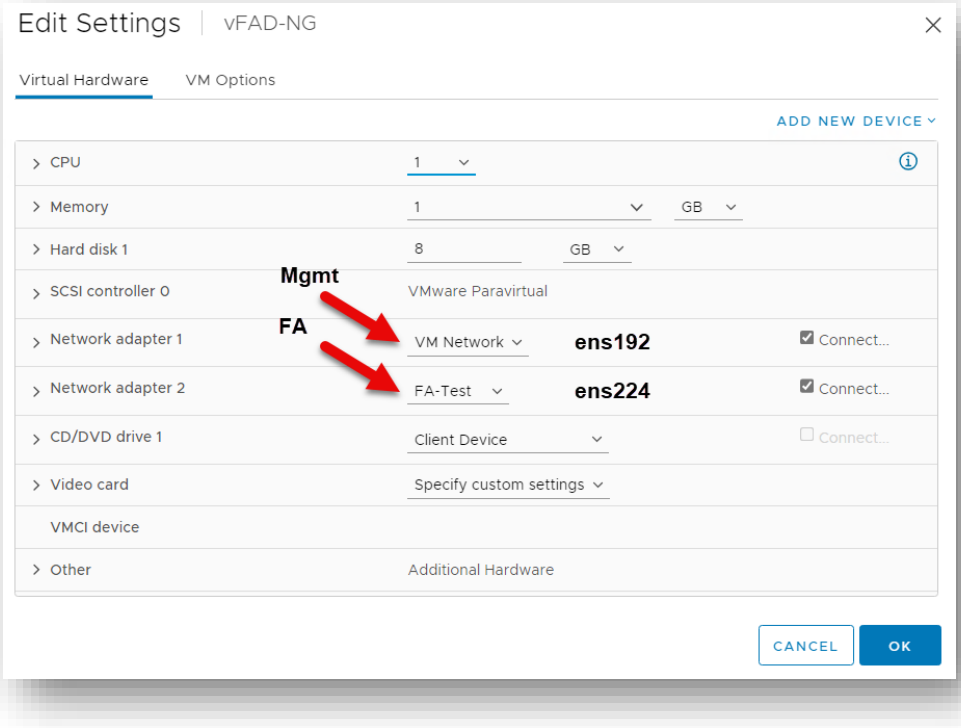
- GNS3 network simulator image
  Loadable image for the network simulator for testing FA functionality in combination with VOSS and EXOS switch simulation.

All options require two physical Ethernet ports: a **Management-NIC** for controlling the device and an **FA-NIC** for providing FA functionality.

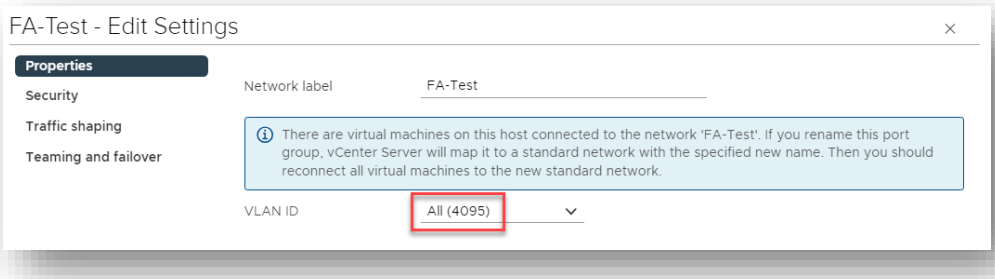
2. vFAD-NG setup

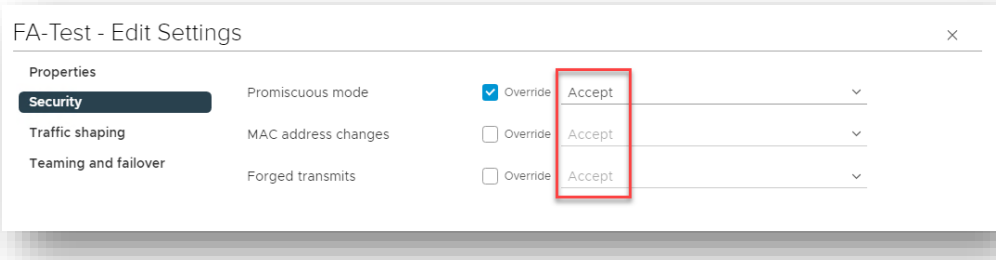
3. Configuring VMware

The VMware Virtual Machine should be configured like this



It is required to manage the vSwitch where **FA** NIC is assigned to, allow promiscuous mode as well as forged transmits otherwise the vFAD will not be able to operate as required.

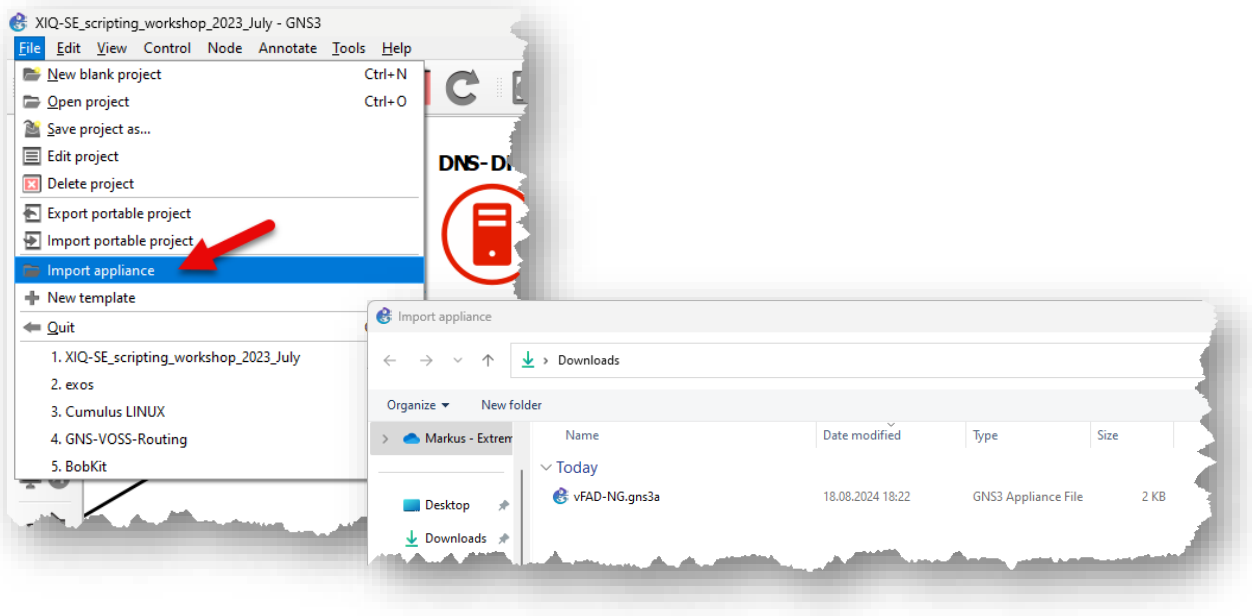


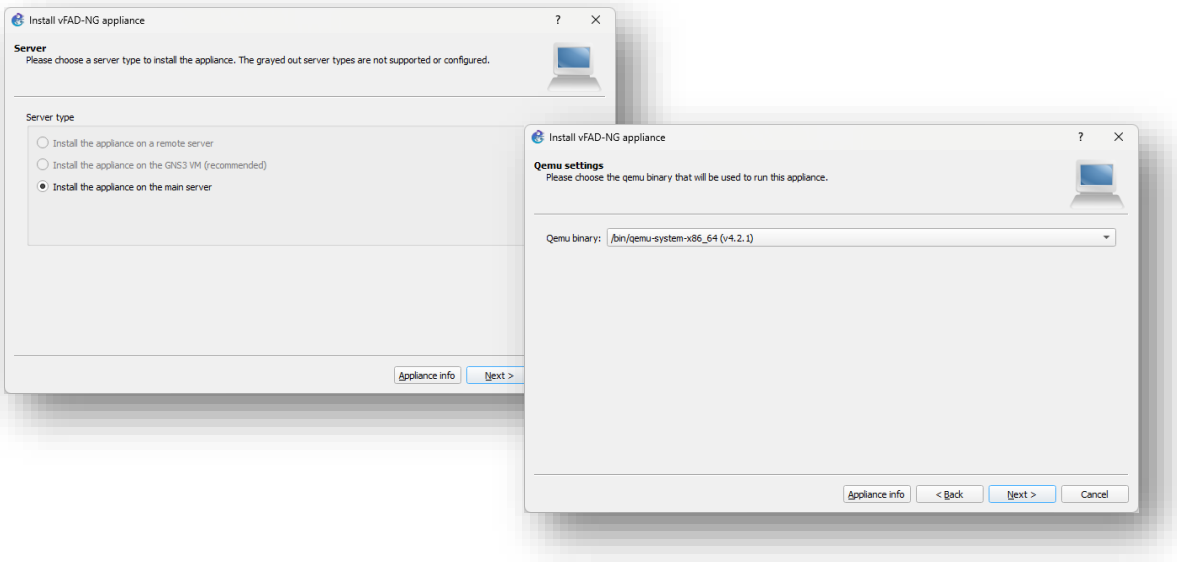


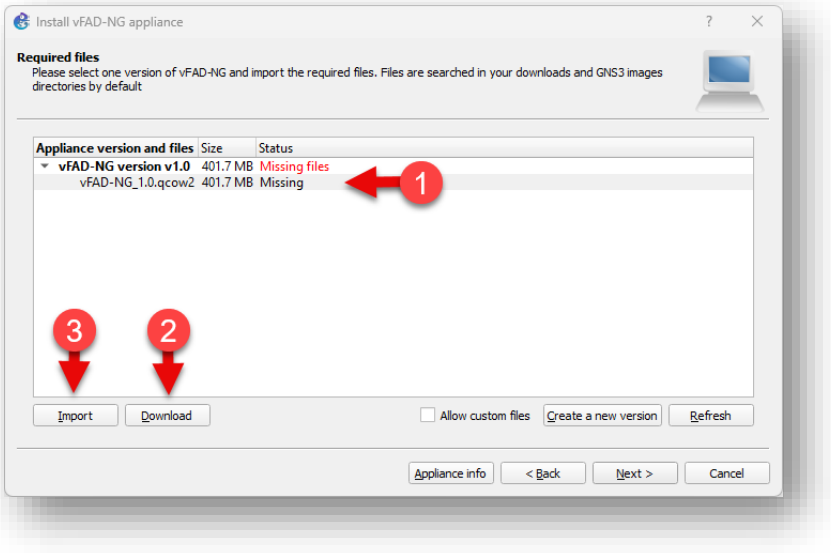
4. Configuring GNS3

The GNS3 network simulator allows simulating network devices and building topologies. Since VOSS 7.0 is it possible to host a SPB fabric in GNS3. With VOSS fabric border nodes and EXOS switches, it is possible to attach a vFAD to simulate VLAN/I-SID signaling. The test was based on GNS3 release 2.2.46.
You can get the GNS3 software from https://www.gns3.com/. You have to download the corresponding vFAD image from the http://nikulski.net/vfad/ website as well. The files have a **.gns3** as well **.qcow2** extension. If you installed the GNS3 management software and deployed a GNS3 VM, you should follow the setup steps to configure the vFAD.

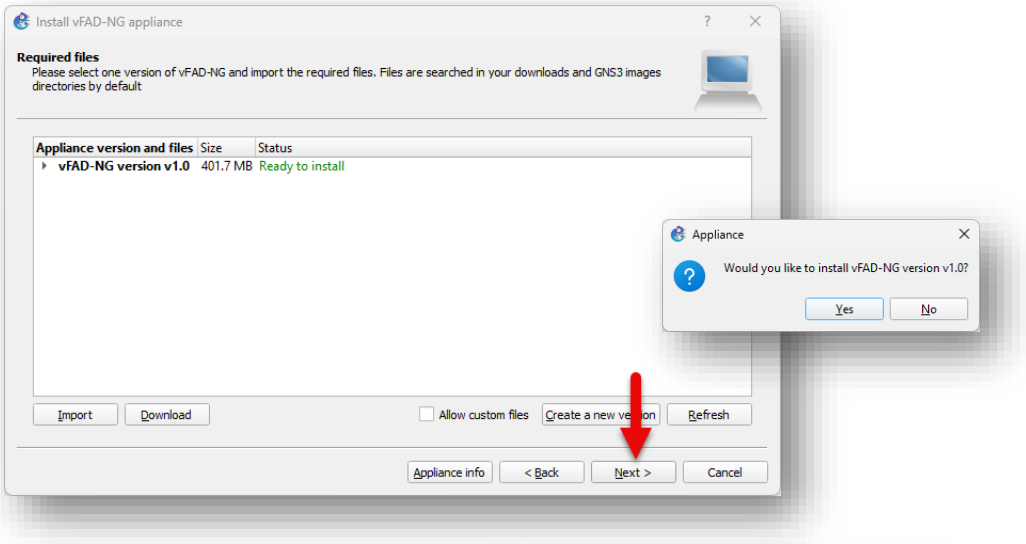
Open your GNS3 client and import the appliance file with the extension **.gns3** like this



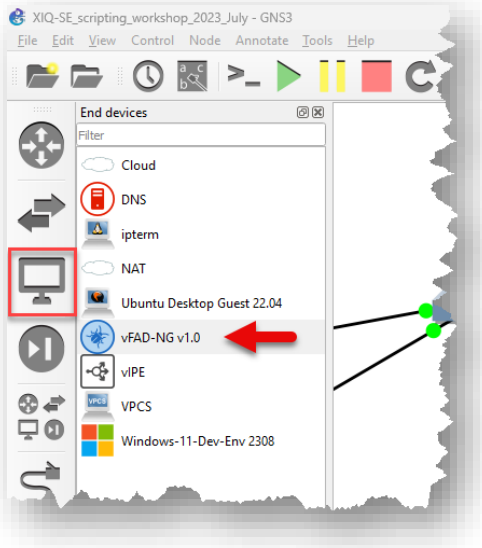




Select the missing qcow2 image file and download (2) if you do not already have the image. If the image is downloaded, your import (3) the file. If the import process is complete, you should now see that the vFAD-NG appliance is ready to install.



Click next to make the appliance available in GNS3 under the end devices section.



5. vFAD-NG Setup

Start the VM. The console and SSH login is **fa** / **fa** .

If required to change to a static IP address, you have to modify the Linux configuration file /etc/network/interfaces. For the yellow part, you must add the **#** sign to deactivate DHCP. The green part: you must remove the leading #, as shown below.

```
sudo -s

nano /etc/network/interfaces

# This file describes the network interfaces available on your system
# and how to activate them. For more information, see interfaces(5).

source /etc/network/interfaces.d/*

# The loopback network interface
auto lo
iface lo inet loopback

# The primary network interface
allow-hotplug ens192
#iface ens192 inet dhcp

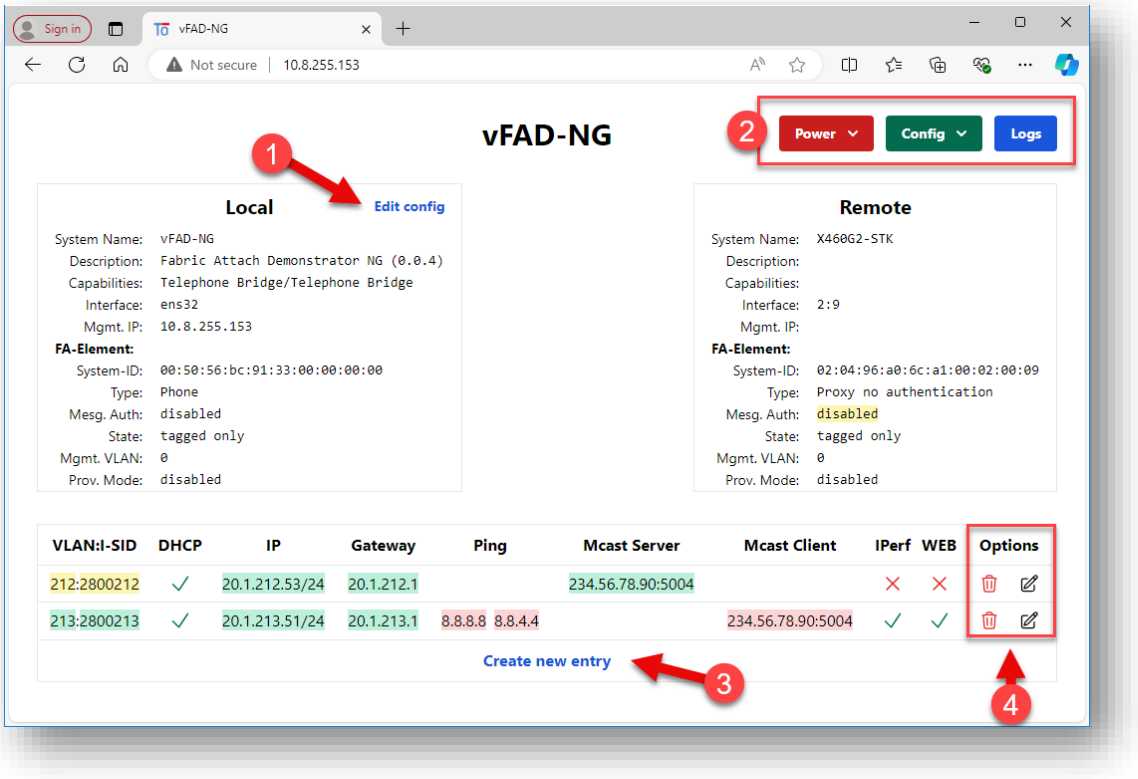
# static IP example  for primary network interface
iface ens192 inet static
        address 1.2.3.4/24
        gateway 1.2.3.1
        dns-nameservers 8.8.8.8
        dns-search example.local

# The secondary network interface
allow-hotplug ens224
iface ens224 inet manual
```

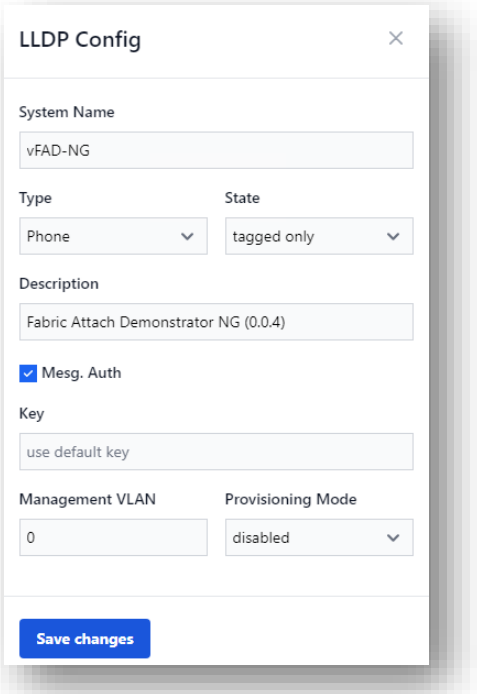
With **CTRL-X** and afterwards **y** and afterwards **ENTER** you can leave the editor. A reboot is required to activate the change.

```
init 6
```

You should now be able to access the management interface by using your preferred WEB browser pointing to the IP address assigned to the Management NIC of the VM using **http** (not https).

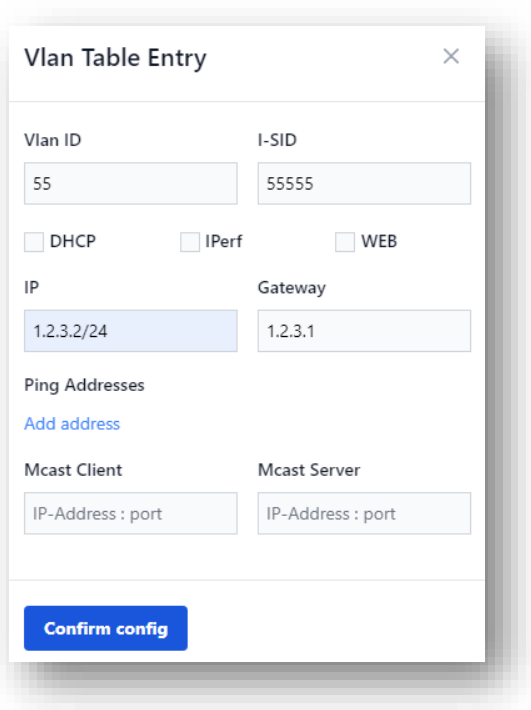


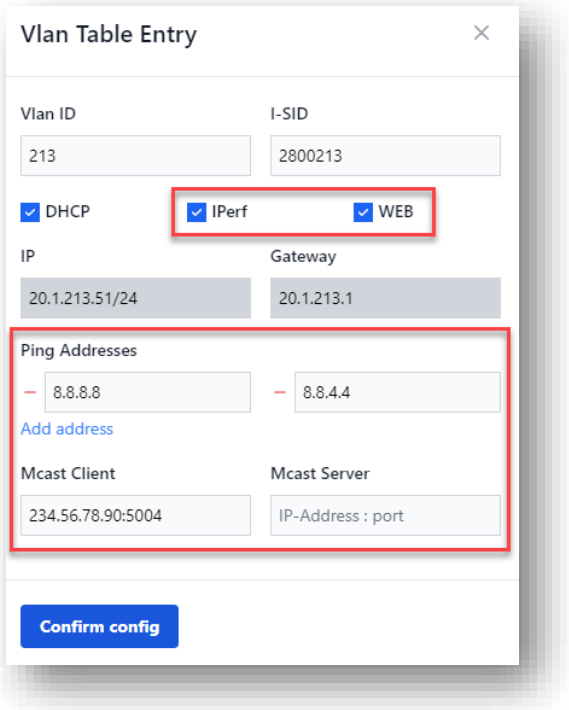
If you click on the left upper area (1), you are able to configure what the vFAD will propagate by **LLDP** and **FA**.



The button block (2) on the right upper corner allows you to control the vFAD appliance.

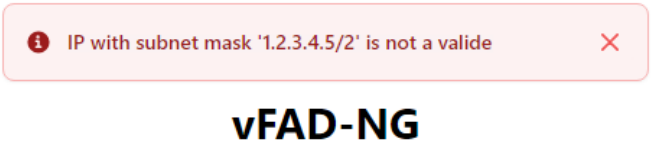
To add an instance of VLAN/I-SID connectivity, click the Create new entry link (3) to open the following dialogue box.

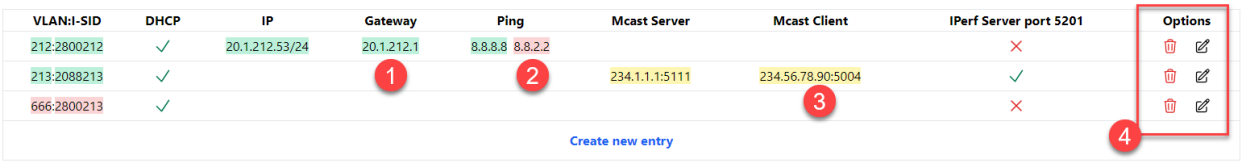




The red angled areas are optional.

In case the input data are malformed or missing, you will see the error message at the top middle like this:

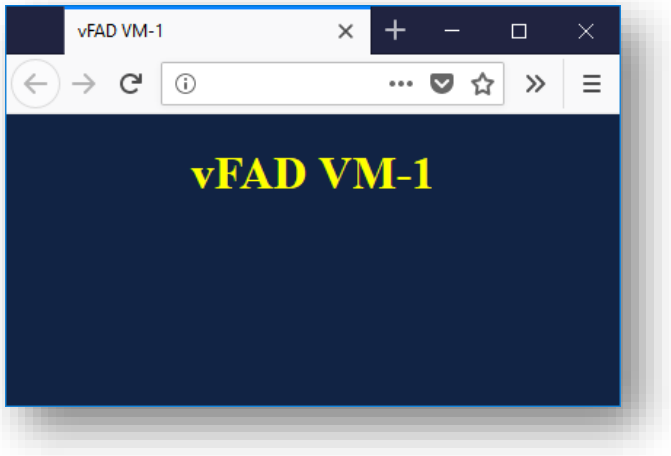




The status is coloured and will be updated multiple times per second. The colours have the following meaning:

212:2800212	accepted
444:444	unknown
666:2800213	rejected
20.1.212.1	is ping able
8.8.2.2	ping timeout
234.56.78.90:5004	wait for multicast stream
234.56.78.90:5004	receive the multicast stream

Each VM instance owns an IP address and a network mask, along with a corresponding default gateway. This IP address can be pinged, and if the WEB server is enabled, you get access to a WEB page like this.



The **IPerf** server works similarly to the WEB server, using the default TCP port. You can use an IPerf3 client like this to generate traffic between your client and the VM instance.

```
iperf3 -c 20.1.212.53
```

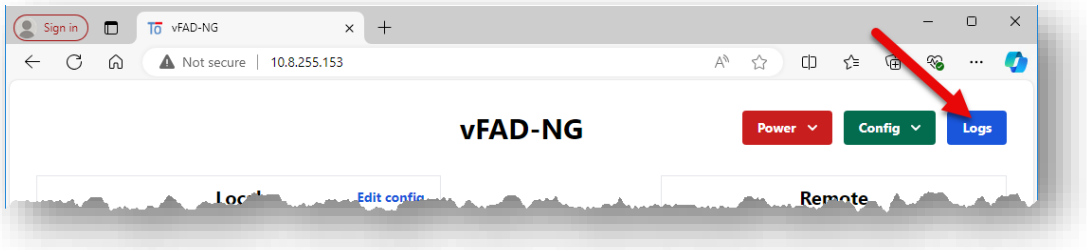
6. Troubleshooting

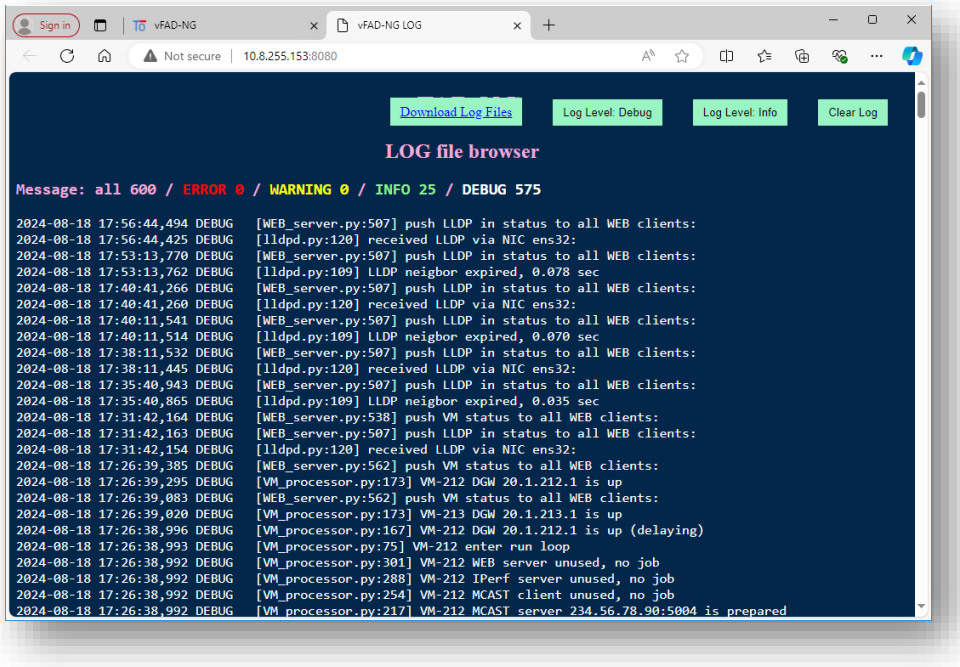
You can also SSH to the same IP with username **fa** and password **fa,** too. Root access isn't enabled via SSH, but you can activate it by logging in first with the fa user and then switching to superuser access by running sudo -s and providing the same **fa** password.

6.1. vFAD

The configuration, status, and LOG information can be accessed via the WEB interface and the CLI. The SSH login is **fa** / **fa**. To grant root access privileges, call **sudo -s** using the same password **fa**.

For troubleshooting the vFAD have a LOG file you can inspect. SSH into the vFAD and call the command log. The default log level is INFO. You can change it to DEBUG. Just call the CLI command debug or info, depending on the LOG level you'd like in place.





Please note that verbose DEBUG LOG messages are not fully displayed. You have to download the log file to access all the details.

1.1 LINUX OS

Through LINUX name space the IP stack is isolated for each VM.

```
ip netns list                                   # all existing
ip netns exec vm-<id> ping -c 1 -W 1 10.0.2.1   # ping from a VM
ip netns exec vm-<id> ip neighbor               # ARP cache from a VM
ip netns exec vm-<id> ip address                # NIC configuration
ip netns exec vm-<id> ip route                  # routes of a VM

htop                                            # show the OS utilization

# capture traffic
tcpdump -i eth1                                 # all traffic
tcpdump -i eth1 arp                             # ARP
tcpdump -i eth1 icmp                            # ping
tcpdump -i eth1 -vv ether proto 0x88cc          # LLDP

debug                                           # enable debug LOG
info                                            # disable debug LOG
tail -n 100 /var/log/vFAD.log                   # logfile
```